

Experiment-10

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Course Code: -MLA0305

Experiment 10 – Advantage Actor-Critic

Title: Implementation of Advantage Actor-Critic (A2C)

Aim:

To develop an Actor-Critic reinforcement learning model using A2C architecture and evaluate its learning efficiency.

Procedure:

1. Select a suitable Reinforcement Learning environment.
2. Create the Actor network for action selection.
3. Create the Critic network for value estimation.
4. Initialize network parameters.
5. Collect states, actions, rewards, and next states.
6. Calculate the advantage function.
7. Update the Actor using the policy gradient.
8. Update the Critic using the value loss.
9. Repeat training over multiple episodes.
10. Plot cumulative reward and learning performance.



```
[2]: import gymnasium as gym, numpy as np, tensorflow as tf
from tensorflow.keras import layers
from tensorflow.keras import layers
env=gym.make('CartPole-v1'); x=keras.Input((4,)); h=layers.Dense(32,activation='relu')(x); out=keras.Model(x,[layers.Dense(2,activation='softmax')(h),layers.Dense(1)(h)]); o
for ep in range(10):
    s,_=env.reset(); S=[]; A=[]; R=[]; done=False
    while not done and len(R)<300:
        p,v=out(np.array([s])); a=np.random.choice(2,p=p.numpy()[0]); ns,r,t,tr,_=env.step(a); S.append(s);A.append(a);R.append(r);s=ns;done=t or tr
    G=[];g=0
    for r in R[::-1]:g=r+.99*g;G.insert(0,g)
    G=np.array(G,dtype=np.float32)
    with tf.GradientTape() as tape:
        p,v=out(np.array(S)); idx=tf.stack([tf.range(len(A)),A],1); lp=tf.math.log(tf.gather_nd(p,idx)+1e-8); adv=G-tf.squeeze(v,1); loss=-tf.reduce_mean(lp*tf.stop_gradient(adv))
    opt.apply_gradients(zip(tape.gradient(loss,out.trainable_variables),out.trainable_variables)); print(ep+1,sum(R))
env.close()
```

1 24.0
2 31.0
3 17.0
4 34.0
5 17.0
6 18.0
7 16.0
8 10.0
9 25.0
10 16.0

Result:

The A2C agent successfully learned to select suitable actions. The Actor improved the policy while the Critic estimated state values, resulting in effective learning.